

Determinant integrals under strong L^n convergence

`MathlibAnnex.MeasureTheory.tendsto_integral_det_of_strongLn`

theorem

Uses Hölder's inequality to turn strong L^n convergence of matrix fields into convergence of their determinant integrals.

Statement

Let $n \geq 1$, let (Ω, μ) be a measure space, and let $\mathcal{F}$ be any filter on an index set I . Let $P_i, Q : \Omega \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$, equipped with the operator norm induced by $\|\cdot\|_\infty$ on $\mathbb{R}^n$. Suppose $Q \in L^n(\mu)$, eventually $P_i \in L^n(\mu)$ along $\mathcal{F}$, and

$$\|P_i - Q\|_{L^n(\mu)} \longrightarrow 0 \quad (i \rightarrow \mathcal{F}).$$

Then $\int_\Omega \det P_i \, d\mu \rightarrow \int_\Omega \det Q \, d\mu$ along the same filter.

Assumptions

The dimension is positive. Membership in L^n includes almost-everywhere strong measurability and finite seminorm. The measure need not be finite, and the filter need not be a sequence filter. No pointwise convergence or common pointwise bound is assumed.

Conclusion

The limiting determinant is integrable, the approximating determinants are eventually integrable, and their integrals converge. In fact their differences converge to zero in $L^1(\mu)$.

The supplied convergence predicate records both seminorm convergence and eventual membership; the separate hypothesis on Q is also necessary for the proof. The one-dimensional case is handled directly, without using the undefined exponent $n/(n-1)$ at $n=1$.

Proof route

Apply the determinant difference estimate, then Hölder with exponents $n/(n-1)$ and n . The resulting bound contains the factor $\|P_i - Q\|_{L^n}$, which tends to zero; the other factor stays bounded by the triangle inequality. Treat $n=1$ directly.

▼ Proof steps

1. **Start with the pointwise estimate and integrability.** The determinant bound gives, at every x ,

$$|\det P_i(x) - \det Q(x)| \leq n(\|P_i(x)\|_{\text{op}} + \|Q(x)\|_{\text{op}})^{n-1} \|P_i(x) - Q(x)\|_{\text{op}}.$$

For any field $T \in L^n(\mu)$, comparison with the zero operator gives

$$|\det T(x)| \leq n\|T(x)\|_{\text{op}}^n, \quad \int_{\Omega} n\|T(x)\|_{\text{op}}^n d\mu < \infty.$$

Continuity of determinant supplies measurability, so $\det T$ is integrable. Apply this to Q and, eventually, to P_i . Hence the determinant difference is eventually integrable. All following estimates are on this eventual set of indices.

2. If $n = 1$, no nontrivial Hölder exponent is needed. Step 1 reduces to

$$\|\det P_i - \det Q\|_{L^1(\mu)} \leq \|P_i - Q\|_{L^1(\mu)} \longrightarrow 0.$$

3. If $n \geq 2$, write Hölder's inequality with the actual functions. Its conjugate exponents are $n/(n-1)$ and n . The functions $x \mapsto \|P_i(x)\|_{\text{op}} + \|Q(x)\|_{\text{op}}$ and $x \mapsto \|P_i(x) - Q(x)\|_{\text{op}}$ belong to $L^n(\mu)$ by the hypotheses and Minkowski's inequality. Thus

$$\begin{aligned} \|\det P_i - \det Q\|_{L^1(\mu)} &\leq n \int_{\Omega} (\|P_i(x)\|_{\text{op}} + \|Q(x)\|_{\text{op}})^{n-1} \|P_i(x) - Q(x)\|_{\text{op}} d\mu \\ &\leq n \left(\int_{\Omega} (\|P_i(x)\|_{\text{op}} + \|Q(x)\|_{\text{op}})^n d\mu \right)^{(n-1)/n} \\ &\quad \cdot \left(\int_{\Omega} \|P_i(x) - Q(x)\|_{\text{op}}^n d\mu \right)^{1/n} \\ &\leq n (\|P_i\|_{L^n(\mu)} + \|Q\|_{L^n(\mu)})^{n-1} \|P_i - Q\|_{L^n(\mu)} \\ &\leq n (\|P_i - Q\|_{L^n(\mu)} + 2\|Q\|_{L^n(\mu)})^{n-1} \|P_i - Q\|_{L^n(\mu)}. \end{aligned}$$

The second inequality is Hölder. The next is Minkowski applied to the two scalar norm functions. The last uses $\|P_i\|_{L^n} \leq \|P_i - Q\|_{L^n} + \|Q\|_{L^n}$. This identifies every function and exponent in the estimate without introducing new names for their norms.

4. Take the limit and then subtract the integrals. Since $\|P_i - Q\|_{L^n(\mu)} \rightarrow 0$, eventually it is at most 1. The last bound in Step 3 is then at most

$$n(1 + 2\|Q\|_{L^n(\mu)})^{n-1} \|P_i - Q\|_{L^n(\mu)} \longrightarrow 0.$$

The coefficient is finite because $Q \in L^n(\mu)$. Together with Step 2 this proves L^1 convergence for every $n \geq 1$. The integrability checked in Step 1 now justifies

$$\begin{aligned} \left| \int_{\Omega} \det P_i d\mu - \int_{\Omega} \det Q d\mu \right| &= \left| \int_{\Omega} (\det P_i - \det Q) d\mu \right| \\ &\leq \|\det P_i - \det Q\|_{L^1(\mu)} \longrightarrow 0. \end{aligned}$$

All eventual statements and limits use the same filter $\mathcal{F}$; neither finite total measure nor a sequence index was assumed.

Main citations

- Exact declaration and its source —
MathlibAnnex.MeasureTheory.tendsto_integral_det_of_strongLn
- Absolute determinant difference bound in the sup operator norm —
MathlibAnnex.ContinuousLinearMap.abs_det_sub_le
- Norm form of the determinant difference bound —
MathlibAnnex.ContinuousLinearMap.norm_det_sub_le
- Strong convergence with eventual dimension-power membership —
MathlibAnnex.MeasureTheory.StrongLnOperatorField

Lean source signature (exact)

```
theorem tendsto_integral_det_of_strongLn
  {n : ℕ} (hn : 0 < n) {α ι : Type*} [MeasurableSpace α]
  {l : Filter ι} {μ : Measure α}
  {P : ι → α → ((Fin n → ℝ) →L[ℝ] (Fin n → ℝ))}
  {Q : α → ((Fin n → ℝ) →L[ℝ] (Fin n → ℝ))}
  (hstrong : StrongLnOperatorField l μ P Q)
  (hQ : MemLp Q n μ) :
  Tendsto (fun i => ∫ x,
    LinearMap.det (P i x : (Fin n → ℝ) →L[ℝ] (Fin n → ℝ)) ∂μ) l
    (ℕ (∫ x,
      LinearMap.det (Q x : (Fin n → ℝ) →L[ℝ] (Fin n → ℝ)) ∂μ))
```

In the source	Mathematical meaning
$\{n : \mathbb{N}\} (hn : 0 < n)$	The positive dimension $n \geq 1$ of the real sup-norm coordinate space.
$\{\alpha \iota : \text{Type}^*\} [\text{MeasurableSpace } \alpha] \{l : \text{Filter } \iota\} \{\mu : \text{Measure } \alpha\}$	The measure space Ω , index set I , arbitrary filter $\mathcal{F}$ on it, and measure μ ; no finiteness or sequence restriction is imposed.
$\{P : \iota \rightarrow \alpha \rightarrow ((\text{Fin } n \rightarrow \mathbb{R}) \rightarrow L[\mathbb{R}] (\text{Fin } n \rightarrow \mathbb{R}))\}$ $\{Q : \alpha \rightarrow ((\text{Fin } n \rightarrow \mathbb{R}) \rightarrow L[\mathbb{R}] (\text{Fin } n \rightarrow \mathbb{R}))\}$	The endomorphism fields $P_i(x), Q(x)$ on Ω , with operator norm induced by the coordinate sup norm.
$(hstrong : \text{StrongLnOperatorField } l \mu P Q)$	The cited definition has two conditions: $\ P_i - Q\ _{L^n(\mu)} \rightarrow 0$ along $\mathcal{F}$, and eventually $P_i \in L^n(\mu)$.
$(hQ : \text{MemLp } Q n \mu)$	The separate assumption $Q \in L^n(\mu)$: a.e. strong measurability and finite L^n seminorm. It is not included in $hstrong$.

In the source	Mathematical meaning
<code>LinearMap.det (P i x : (Fin n → ℝ) →₁[ℝ] (Fin n → ℝ))</code>	The real signed determinant $\det P_i(x)$; the cast keeps the same map. The analogous expression denotes $\det Q(x)$.
<code>Tendsto (fun i => ∫ x, LinearMap.det (P i x : (Fin n → ℝ) →₁[ℝ] (Fin n → ℝ)) ∂μ) 1</code>	The conclusion sends $\int_{\Omega} \det P_i d\mu$ to $\int_{\Omega} \det Q d\mu$ along the same filter $\mathcal{F}$, as specified by the final neighbourhood expression.

▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section

open Set MeasureTheory Filter
open scoped BigOperators ENNReal NNReal Topology

namespace MathlibAnnex.MeasureTheory

```

▼ Exact content identity

Declaration: `MathlibAnnex.MeasureTheory.tendsto_integral_det_of_strongLn`
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Accepted source guide SHA-256: `e3f0a706712505e7e4abc489e7f2b514f24be3fe57ac5a1956708d00ac984e9e`
Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`